# Sanaul Haque

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# IARCO RESEARCH PROPOSAL

**Full Legal Name:** Aiana Zhagparova

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**Category:** Junior

**Class/Grade/Year:** 12 Grade

**Country:** Kazakhstan

**Submission Date:** September 20, 2026

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**Research Topic:** Evaluating a Pharmacogenetic-Guided Warfarin Dosing Approach in Adults with Atrial Fibrillation in Kazakhstan

Title: Evaluating a Pharmacogenetic-Guided Warfarin Dosing Approach in Adults with Atrial Fibrillation in Kazakhstan

**Research Problem**

Warfarin remains an essential anticoagulant for patients with atrial fibrillation (AF) who have mechanical heart valves or moderate-to-severe rheumatic mitral stenosis, or for whom direct oral anticoagulants are unsuitable [4]. However, it has a narrow therapeutic index, and the dose required varies considerably between patients. In the United States, warfarin was implicated in 33.3% of emergency hospitalizations for adverse drug events among adults aged 65 and older, more than any other medication [1]. Variants in *CYP2C9* and *VKORC1*, together with nongenetic factors, explain roughly 50% of the variability in dose requirements [2]. Allele frequencies also differ between populations. In Kazakhs, the *VKORC1* rs8050894 G allele frequency is 63%, compared with about 94% in Asian and 37% in Caucasian reference populations, and *CYP2C9*2* and **3* frequencies are 5% and 7% [3]. Most dosing algorithms were developed in European, East Asian, and African-American populations [5], so it is unknown whether they perform equally well in Kazakh patients. This research seeks to evaluate whether genotype-guided warfarin dosing improves anticoagulation control in Kazakhstan, contributing evidence for clinical practice and health policy.

**Existing Literature**

Pharmacogenetic information is established as a biological determinant of warfarin response. *CYP2C9* variants reduce S-warfarin metabolism, *VKORC1* -1639G>A increases sensitivity to the drug, and *CYP4F2* V433M modestly raises dose requirements [2], [5], [6]. Clinical trials, however, disagree. EU-PACT (n = 455) reported 67.4% time in therapeutic range (TTR) with genotype-guided dosing versus 60.3% with standard dosing [7]. COAG (n = 1,015) found no difference at four weeks (45.2% vs. 45.4%) [8]. A meta-analysis of 10 randomized trials found no improvement in the first month but a TTR increase from 53.0% to 59.4% afterward, with lower major bleeding [9]. Another meta-analysis reported benefit but substantial heterogeneity ($I^2 = 80\%$) [10]. Recent work suggests feasibility in lower-resource settings: a randomized trial in India (n = 168) showed faster attainment of therapeutic INR and greater TTR [11], and point-of-care testing in three UK clinics raised TTR from 55.25% to 62.74% [12]. In Kazakhstan, earlier work developed a warfarin genotyping approach and testing kit [13], but no prospective clinical evaluation of genotype-guided dosing has been reported. Economic evidence is also mixed: one model found testing unlikely to be cost-effective for typical nonvalvular AF patients [14], while an EU-PACT-based analysis found it potentially cost-effective in the UK and Sweden [15]. This study fills the gap by combining clinical, genetic, economic, and implementation data in a Kazakhstan-specific evaluation.

**Research Question**

To what extent does genotype-guided warfarin dosing (*CYP2C9*2*, *CYP2C9*3*, *VKORC1* -1639G>A) improve anticoagulation control in patients with AF in Kazakhstan, in terms of time in therapeutic INR range, time to stable therapeutic INR, and bleeding and thromboembolic events, and is it feasible and economically justified in the local healthcare system?

## Methodology

This study adopts a mixed-methods approach to evaluate both clinical effectiveness and practical implementation.

Quantitative Component:

- A prospective pilot study in one or two major cardiology centers will compare patients starting warfarin under genotype-guided dosing with a control group receiving standard clinical dosing.

- Genotyping for *CYP2C9*2* (rs1799853), *CYP2C9*3* (rs1057910), and *VKORC1* -1639G>A (rs9923231) will be performed before or at treatment initiation. Doses will be estimated with a validated pharmacogenetic algorithm [5] that also uses age, body size, and interacting medications. The dose will serve as an initial estimate, with INR monitoring guiding all adjustments.

- Participants will be followed for 90 days. Sample size will be finalized by power calculation; a target of about 120 patients (60 per group) is proposed for the pilot.

- Primary outcome: percentage of time in therapeutic INR range. Secondary outcomes: time to first therapeutic INR, INR values ≥ 4.0, major bleeding, thromboembolic events, and testing turnaround time.

- Allele frequencies in the study cohort will be compared with those reported in [3]. Group differences will be analyzed with appropriate parametric or non-parametric tests, with adjustment for age, comorbidities, and interacting drugs.

- An economic evaluation will estimate testing costs, INR monitoring, and costs of bleeding and thromboembolic complications from the healthcare-system perspective.

Qualitative Component:

- Semi-structured interviews with about 10 cardiologists and clinic physicians, 5 laboratory staff, and 10 patients will explore feasibility, acceptability, and barriers to implementation.

- Thematic analysis will be used to extract insights, using NVivo software.

Future Extension:

- If the pilot generates a sufficiently large local dataset, machine-learning dosing models could later be evaluated against the validated algorithm. AI is not part of the initial intervention.

1 Ethical Considerations:

- Ethical approval will be obtained from a recognized institutional review board.

- Informed consent will be obtained from all participants, including consent for genetic testing.

1 - Genetic data confidentiality and anonymity will be strictly maintained.

**Research Topic**

This topic is timely and locally relevant. Warfarin remains in use despite the rise of DOACs, and Kazakh-specific genetic data show allele frequencies that differ from reference populations. Rather than assuming that international algorithms transfer, this research evaluates them prospectively within the existing treatment pathway. By combining clinical outcomes, economic assessment, and stakeholder perspectives, it offers useful insights to clinicians, healthcare policymakers, and the wider pharmacogenetics community.

1 **Quality of Writing**

The language of this proposal is academically precise, clear, and appropriate for a multidisciplinary audience. Jargon is minimized, and the structure follows standard research norms. The proposal adheres to grammar, citation, and formatting guidelines to meet IARCO standards.

## References


[1] D. S. Budnitz et al., "Emergency hospitalizations for adverse drug events in older Americans," *N. Engl. J. Med.*, vol. 365, no. 21, pp. 2002–2012, 2011. DOI: 10.1056/NEJMsa1103053.

[2] J. A. Johnson et al., "Clinical Pharmacogenetics Implementation Consortium guidelines for CYP2C9 and VKORC1 genotypes and warfarin dosing," *Clin. Pharmacol. Ther.*, vol. 90, no. 4, pp. 625–629, 2011. DOI: 10.1038/clpt.2011.185.

[3] A. N. Iskakova et al., "Polymorphisms in genes involved in the absorption, distribution, metabolism, and excretion of drugs in the Kazakhs of Kazakhstan," *BMC Genet.*, vol. 17, p. 23, 2016. DOI: 10.1186/s12863-016-0329-x.

[4] J. A. Joglar et al., "2023 ACC/AHA/ACCP/HRS guideline for the diagnosis and management of atrial fibrillation," *Circulation*, vol. 149, pp. e1–e156, 2024. DOI: 10.1161/CIR.0000000000001193.

[5] J. A. Johnson et al., "CPIC guideline for pharmacogenetics-guided warfarin dosing: 2017 update," *Clin. Pharmacol. Ther.*, vol. 102, no. 3, pp. 397–404, 2017. DOI: 10.1002/cpt.668.

[6] M. D. Caldwell et al., "CYP4F2 genetic variant alters required warfarin dose," *Blood*, vol. 111, no. 8, pp. 4106–4112, 2008. DOI: 10.1182/blood-2007-11-122010.

[7] M. Pirmohamed et al., "A randomized trial of genotype-guided dosing of warfarin," *N. Engl. J. Med.*, vol. 369, no. 24, pp. 2294–2303, 2013. DOI: 10.1056/NEJMoa1311386.

[8] S. E. Kimmel et al., "A pharmacogenetic versus a clinical algorithm for warfarin dosing," *N. Engl. J. Med.*, vol. 369, no. 24, pp. 2283–2293, 2013. DOI: 10.1056/NEJMoa1310669.

[9] K. Dahal et al., "Meta-analysis of randomized controlled trials of genotype-guided vs standard dosing of warfarin," *Chest*, vol. 148, no. 3, pp. 701–710, 2015. DOI: 10.1378/chest.14-2947.

[10] R. Goulding et al., "Genotype-guided drug prescribing: A systematic review and meta-analysis of randomized control trials," *Br. J. Clin. Pharmacol.*, vol. 80, no. 4, pp. 868–877, 2015. DOI: 10.1111/bcp.12672.

[11] A. Anand et al., "Pharmacogenetic guided versus standard warfarin dosing for routine clinical care with its pharmacoeconomic impact: A randomized controlled clinical trial," *Ann. Hematol.*, vol. 103, no. 6, pp. 2133–2144, 2024. DOI: 10.1007/s00277-024-05757-1.

[12] A. L. Jorgensen et al., "Implementation of genotype-guided dosing of warfarin with point-of-care genetic testing in three UK clinics: A matched cohort study," *BMC Med.*, vol. 17, p. 76, 2019. DOI: 10.1186/s12916-019-1308-7.

[13] E. Zholdybayeva et al., "Pharmacogenetic research in Kazakhstan," *Cent. Asian J. Glob. Health*, vol. 2, p. 87, 2014. DOI: 10.5195/cajgh.2013.87.

[14] M. H. Eckman et al., "Cost-effectiveness of using pharmacogenetic information in warfarin dosing for patients with nonvalvular atrial fibrillation," *Ann. Intern. Med.*, vol. 150, no. 2, pp. 73–83, 2009. DOI: 10.7326/0003-4819-150-2-200901200-00005.

[15] T. I. Verhoef et al., "Cost-effectiveness of pharmacogenetic-guided dosing of warfarin in the United Kingdom and Sweden," *Pharmacogenomics J.*, vol. 16, no. 5, pp. 478–484, 2016. DOI: 10.1038/tpj.2016.41.